

LAS CRUCES 20 N, NM, USA

WMO: 746320

Lat: 32.6137N

Lon: 106.7414W

Elev: 4327

StdP: 12.54

Time Zone: -7.00 (NAM)

Period: 07-19

WBAN: 03074

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR							Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%				0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD		
(a) 1	(b) 12.5	(c) 16.4	(d) -6.7	(e) 4.5	(f) 39.8	(g) -2.8	(h) 5.5	(i) 41.2	(j) 20.7	(k) 54.4	(l) 18.3	(m) 55.6	(n) 1.5	(o) N/A	(p) 0.430	

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 26.4	(c) 99.2	(d) 60.5	(e) 97.0	(f) 60.6	(g) 95.1	(h) 60.8	(i) 68.1	(j) 79.1	(k) 67.2	(l) 79.5	(m) 66.3	(n) 79.7	(o) 6.6	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.8	112.1	70.0	64.4	106.3	70.0	63.0	101.2	70.4	35.0	78.2	34.2	79.2	33.6	79.1	72.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years						
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max					
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.4	15.5	13.0	DB	4.3	104.0	8.0	1.5	-1.5	105.0	-6.2	105.9	-10.6	106.7	-16.5	107.8	
			WB	2.2	70.3	8.8	1.3	-4.1	71.2	-9.3	72.0	-14.2	72.7	-20.6	73.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	59.8	39.1	44.5	52.1	58.8	66.2	77.8	79.7	78.1	71.8	60.5	48.2	40.5
	DBStd	15.62	6.83	8.11	6.89	6.40	5.83	4.52	3.35	3.39	4.46	6.93	7.36	7.60
	HDD50	1028	343	181	61	9	1	0	0	0	0	8	120	305
	HDD65	3472	803	574	400	202	53	0	0	0	6	170	505	759
	CDD50	4623	5	28	126	272	504	833	919	871	653	335	66	11
	CDD65	1590	0	0	1	15	92	383	454	406	208	31	0	0
	CDH74	23540	0	23	221	796	2397	6308	5586	4821	2574	780	34	0
	CDH80	11840	0	0	22	177	1013	3772	3052	2519	1098	187	0	0
Wind	WSAvg	4.5	3.6	4.8	5.3	6.5	5.9	5.2	4.3	3.8	3.6	3.8	3.6	3.8
Precipitation	PrecAvg	10.6	0.5	0.4	0.3	0.2	0.4	0.8	2.0	2.2	1.4	0.9	0.6	0.8
	PrecMax	20.0	1.7	2.1	1.1	2.4	3.6	4.5	5.3	6.6	3.7	3.4	3.1	3.9
	PrecMin	5.1	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.2	0.0	0.0	0.0	0.0
	PrecStd	3.4	0.4	0.4	0.3	0.4	0.6	0.9	1.3	1.2	1.0	0.8	0.7	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.6	77.3	82.9	88.0	95.1	103.0	101.0	98.8	94.7	88.0	77.7	69.2
		MCWB	45.7	49.4	50.6	52.9	55.7	59.4	61.6	63.2	60.9	57.0	51.4	47.7
	2%	DB	64.2	73.0	79.5	84.5	92.2	100.3	98.3	96.5	91.6	84.9	74.4	65.5
		MCWB	44.3	47.4	49.4	51.9	55.1	59.2	62.0	63.2	61.1	56.7	50.5	46.2
	5%	DB	60.8	68.8	76.5	81.9	89.1	98.0	96.1	94.6	89.3	81.9	70.9	61.2
		MCWB	42.7	45.7	48.1	51.1	54.2	58.8	62.3	63.4	60.5	55.4	50.1	44.7
	10%	DB	56.9	64.3	73.0	79.1	86.0	95.7	93.7	91.9	86.8	78.7	66.6	57.2
		MCWB	40.9	44.2	46.7	50.1	52.9	58.6	62.6	63.5	60.4	54.8	48.0	42.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.8	51.5	52.8	56.8	60.6	65.6	69.3	69.3	67.6	65.6	56.5	50.5
		MCDB	61.5	64.3	71.7	78.3	83.0	85.5	78.0	79.2	75.6	76.5	67.0	62.8
	2%	WB	46.1	49.4	50.7	54.5	58.5	64.2	68.2	68.0	66.4	62.3	54.1	48.2
		MCDB	59.0	67.7	73.4	77.0	81.4	85.4	78.5	80.2	76.2	72.8	64.0	61.9
	5%	WB	44.1	47.5	49.2	52.6	56.5	63.2	67.4	67.1	65.5	59.2	52.1	46.1
		MCDB	57.7	65.8	73.1	76.5	80.3	85.8	79.0	80.7	76.5	73.5	64.5	58.7
	10%	WB	42.1	45.3	47.8	51.1	54.7	62.0	66.5	66.3	64.3	57.3	49.7	43.9
		MCDB	55.3	63.4	70.7	76.3	80.4	86.6	79.5	80.8	76.9	72.7	64.2	55.9
Mean Daily Temperature Range	5% DB	MDBR	33.4	35.4	36.2	35.9	35.3	34.4	26.4	26.7	27.7	32.9	33.7	30.0
		MCDBR	38.8	42.3	42.1	40.0	40.3	38.4	31.1	30.9	32.3	36.3	38.1	34.9
		MCWBR	22.2	21.8	20.6	18.5	17.1	14.6	8.8	7.7	10.8	15.7	18.7	20.1
	5% WB	MCDBR	30.9	35.5	37.5	35.7	34.1	29.4	23.5	24.7	21.2	25.6	27.4	29.4
MCWBR		17.8	18.8	18.6	16.7	14.5	10.1	5.8	6.6	7.4	11.4	14.8	17.1	
Clear-Sky Solar Irradiance	taub		0.242	0.247	0.266	0.288	0.307	0.330	0.368	0.347	0.317	0.275	0.255	0.244
	taud		2.664	2.654	2.558	2.469	2.465	2.425	2.407	2.443	2.505	2.623	2.627	2.651
	Ebn at Noon		319	326	324	318	310	301	289	295	302	311	311	313
	Edh at Noon		23	26	30	35	36	37	37	36	32	26	23	22
All-Sky Solar Radiation	RadAvg		1128	1446	1872	2296	2519	2525	2256	2088	1861	1544	1213	1019
	RadStd		78	113	99	91	85	111	108	73	85	116	73	55

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (11 neighbors)	+0.72	N/A	N/A	N/A	N/A	N/A	N/A	-160	+263	+149